$\Rightarrow x - 5 = 2x - 30$

$\Rightarrow x = 25.$

Present age of the elder man = $(x + 10)$ years = $(25 + 10)$ years = 35 years.

46. Let the ages of Kunal and Sagar 6 years ago be $6x$ and $5x$ years.

Then, $\dfrac{(6x+6)+4}{(5x+6)+4} = \dfrac{11}{10} \Rightarrow \dfrac{6x+10}{5x+10} = \dfrac{11}{10}$

$\Rightarrow 10\ (6x + 10) = 11\ (5x + 10)$

$\Rightarrow 60x + 100 = 55x + 110$

$\Rightarrow 5x = 10$

$\Rightarrow x = 2.$

Sagar's present age = $(5x + 6)$ years = $(5 \times 2 + 6)$ years = 16 years.

47. Vimal's present age = $(8 + 2)$ years = 10 years.

Sneh's father's age = 2 $(10 + 10)$ years = 40 years.

Sneh's age = $\left(\dfrac{1}{6}\times 40\right)$ years $= \dfrac{20}{3}$ years $= 6\dfrac{2}{3}$ years.

48. Let Samina's age be $7x$ years. Then, Suhana's age = $3x$ years.

$\therefore \dfrac{7x+6}{3x+6} = \dfrac{5}{3} \Rightarrow 3\ (7x + 6) = 5\ (3x + 6)$

$\Rightarrow 21x + 18 = 15x + 30$

$\Rightarrow 6x = 12$

$\Rightarrow x = 2.$

Difference in their ages = $(7x - 3x)$ years = $4x$ years = (4×2) years = 8 years.

49. Let Sulekha's age be $9x$ years. Then, Arunima's age = $8x$ years.

$\dfrac{9x+5}{8x+5} = \dfrac{10}{9} \Rightarrow 9\ (9x + 5) = 10\ (8x + 5)$

$\Rightarrow 81x + 45 = 80x + 50$

$\Rightarrow x = 5.$

Difference in their ages = $(9x - 8x)$ years = x years = 5 years.

50. Let Amisha's age 3 years ago be $8x$ years. Then, Nimisha's age 3 years ago = $9x$ years.

Present age of Amisha = $(8x + 3)$ years.

Present age of Nimisha = $(9x + 3)$ years.

$\dfrac{(8x+3)+3}{(9x+3)+3} = \dfrac{11}{12} \Rightarrow \dfrac{8x+6}{9x+6} = \dfrac{11}{12}$

$\Rightarrow 12\ (8x + 6) = 11\ (9x + 6)$

$\Rightarrow 96x + 72 = 99x + 66$

$\Rightarrow 3x = 6$

$\Rightarrow x = 2.$

Amisha's present age = $(8 \times 2 + 3)$ years = 19 years.

51. Saloni's age = 14 years $\Rightarrow$ Sachin's age = $(14 - 9)$ years = 5 years.

Let the present age of Mr. Roy be x years.

$\dfrac{x-10}{14} = 5 \Rightarrow x - 10 = 70 \Rightarrow x = 80$ years.

52. Let the present age of Y be a years.

Three years ago X's age = $3a$ years

Then, present age of X is $(3a + 3)$

Z's present age = $2a$

According to the given information

Now, $(3a + 3) - 2a = 12 \Rightarrow a = 9$ year

$\therefore$ Present age of $Z = 2a = 2 \times 9 = 18$ years

53. Let the age of the son and the daughter of Poorvi be $6a$ years and $7a$ years respectively. 5 years hence, present age of son = $6a - 5$ and present age of daughter = $7a - 5$

According to the question,

Eight years ago, the age of Poorvi $= 6a - 5 + 7a - 5$
$= 13a - 10$

So, present age of Poorvi = $13a - 10 + 8 = 13a - 2.$

Since, present age of Poorvi husband = $3(6a - 5)$

The difference of present age of Poorvi husband and Poorvi = 7, (given)

$3(6a-5)-(13a-2)=7$, $\Rightarrow 18a - 15 - 13a + 2 = 7$

$\Rightarrow 5a = 20 \quad \Rightarrow a = 4$

The present age of daughter = $(7a - 5) = 7 \times 4 - 5 = 23$ years

54. Let present age of father, mother and son be x, y and z respectively

Sum of present ages of father and son = (Mother's present age + 8 years)

$\Rightarrow x + z = y + 8$ years ...(i)

Mother's present age = (Son's present age + 22)

$\Rightarrow y = z + 22$ ...(ii)

Put the value of y in equation (i) we get

$x + z = z + 22 + 8$

$\Rightarrow x + z = z + 30$

$\Rightarrow x = 30$ years

$\therefore$ Father's present age = 30 years

Age of father after four years = 30 + 4 = 34 years

$\therefore$ Required age of father

= 34 years

55. Let the age of Rahul, Sagar and Purav be x, y and z respectively

According to the given information

Age of Sagar – Age of Rahul = Age of Rahul – Age of Purav

$\Rightarrow y - x = x - z$

$\Rightarrow 2x = y + z$ ...(i)

Also $y + z = 66$ years

From (i) $x = 33$ years

Also as per Eq (i) we have Purav's age + Sagar's age = 66 years.

By going through option (a) given Purav = 18, and Rahul = 33 years, Sagar = 48 years

Difference between Rahul's and Purav's age = 18 years

56. Let the present age of A be a years and that of B be b years.

Then, 4 years ago,

A's age = $(a - 4)$ years

B's age = $(b - 4)$ years